PAMANTASAN NG LUNGSOD NG VALENZUELA
College of Engineering and Information Technology
INFORMATION TECHNOLOGY DEPARTMENT

Design and Development of iDetect: An OCR-Based Discount ID Verification Application for Business Establishments under the Business Permit and Licensing Office in Valenzuela City

An Undergraduate Capstone Project Presented to the Faculty of

College of Engineering and Information Technology Department

Pamantasan ng Lungsod ng Valenzuela

Valenzuela City

In Partial Fulfillment of the Requirements for the Degree

Bachelor of Science in Information Technology

Presented by:

DE LOS REYES, CARLO M.

ENORIA, JORICH MARICAR C.

MERCADO, VESTI SAHM ANGIELO

SANTIAGO, ISHMAEL LUIS C.

February 2026

Functional Requirements

1.1 System Infrastructure and Environment Setup

Description and Priority

This foundational feature covers the establishment of the project development environment, including the project repository, database schemas, API scaffolding, and all development tooling required before any portal feature can be built or tested. All subsequent development tasks across the BPLO, OSCA, PDAO, Client Portal, and Beneficiary Mobile Application depend on a stable and verified baseline environment.

Priority: High

Stimulus / Response

Positive Flow

- The system administrator initializes the project repository and configures the required development tooling (e.g., version control, CI/CD pipelines, linters).

  Response: The repository is successfully created, all team members are granted appropriate access, and the development environment is confirmed operational.

- The developer runs the database migration scripts to set up the initial database schemas.

  Response: All database schemas are successfully created and verified, with seed data populated where applicable.

- The developer deploys the API scaffolding and runs a health check.

Response: The API endpoints respond correctly, confirming that the base infrastructure is stable and ready for feature development across all portals (BPLO, OSCA, PDAO, Client Portal, and Beneficiary Mobile Application).

Negative Flow

- The system administrator attempts to initialize the repository but encounters a configuration or permission error.

  Response: The system notifies the administrator of the specific error, and repository setup is halted until the issue is resolved.

- The developer runs the database migration scripts but encounters a schema conflict or connection failure.

  Response: The system rolls back any partial migrations, logs the error details, and notifies the developer that environment setup cannot proceed until the database issue is resolved.

Functional Requirements

REQ - 1.1.1: The system development environment shall be configured with a project repository, development tools, mobile and web frameworks, database instances, and API scaffolding verified and operational before any portal development task begins, providing a working baseline for all subsequent features.

REQ - 1.1.2: All database tables, relationships, and initial seed data for OSCA, PDAO, BPLO, establishment, and beneficiary records shall be defined, along with the

RESTful API endpoint structure across all five portals, as part of the database schema and API scaffolding setup.

1.2. Admin Account Pre-Configuration

Description and Priority

This feature covers the pre-configuration of administrator accounts for BPLO, OSCA, and PDAO officers prior to system deployment. Each office manages its own personnel credentials through its respective portal, and no cross-portal administrative access is permitted.

Priority: High

Stimulus / Response

Positive Flow

- A system administrator creates and pre-configures an admin account for a BPLO, OSCA, or PDAO officer prior to system deployment.

  Response: The account is successfully created within the respective office portal, credentials are generated, and the officer is notified with their login details.

- An office administrator logs into their respective portal using the pre-configured credentials.

  Response: The system authenticates the officer and grants access exclusively to their assigned portal (BPLO, OSCA, or PDAO), with no visibility or access to other portals.

- An office administrator manages personnel credentials within their own portal (e.g., adding or updating staff accounts).

Response: The system saves the changes and reflects them within the office's portal only, maintaining data isolation across BPLO, OSCA, and PDAO.

Negative Flow

- A system administrator attempts to create an admin account with incomplete or invalid credentials (e.g., duplicate username, missing required fields).

  Response: The system rejects the account creation, displays a specific validation error, and prompts the administrator to correct the input before resubmitting.

- An officer attempts to log in using incorrect or expired credentials.

  Response: The system denies access, displays an authentication error, and prompts the officer to verify their credentials or contact their system administrator.

Functional Requirements

REQ - 1.2.1: BPLO Administrator accounts must be created and verified with correct portal access permissions before establishment account provisioning can begin.

REQ - 1.2.2: OSCA and PDAO Administrator accounts must be created and verified with correct portal access permissions, ensuring each office manages only its own records with no cross-portal access.

2.1. Logging In and Authenticating Users

Description and Priority

This feature allows all system users: establishment staff, BPLO officers, OSCA administrators, PDAO administrators, and beneficiaries to securely log in to their respective portals using role-specific credentials. Two-factor authentication is enforced for beneficiary accounts to balance security with accessibility.

Priority: High

Stimulus / Response

Positive Flow

- An establishment staff enters their registered username and password in the Client Portal login screen.

  Response: The system shall authenticate the credentials, verify the establishment's active business permit status, and grant access to the verification dashboard.

- A beneficiary enters their ID number and birthdate, then submits the OTP sent to their registered mobile number.

  Response: The system shall validate both authentication factors against the OSCA/PDAO database records and grant access to the Beneficiary Mobile Application.

- A BPLO, OSCA, or PDAO administrator enters their institutionally issued credentials in their respective portal login screen.

Response: The system shall authenticate the credentials against the administrator account database, verify the account is active, and grant access to the appropriate administrative portal dashboard.

Negative Flow

- A user enters incorrect login credentials three consecutive times.

  Response: The system shall temporarily lock the account for a defined cooling-off period and display a lockout notification with instructions to contact their administrator.

- A beneficiary enters a correct ID number and birthdate but submits an expired or incorrect OTP.

  Response: The system shall deny access and prompt the beneficiary to request a new OTP, with a maximum resend limit per session.

Functional Requirements

REQ - 2.1.1: The system shall authenticate establishment staff credentials by verifying the username and password against the account database, and confirm that the associated establishment account has an approved and active status within the Client Portal.

REQ - 2.1.2: The system shall implement two-factor authentication for beneficiaries by requiring their ID number paired with their birthdate as the primary factor, and a one-time PIN delivered to their registered mobile number as the secondary factor within the login workflow.

REQ - 2.1.3: The system shall pre-configure administrator accounts for BPLO, OSCA, and PDAO officers prior to system deployment, with each office responsible for managing its own personnel credentials through its respective portal, and no cross-portal administrative access permitted.

REQ - 2.1.4: The system shall temporarily lock a user account after three consecutive failed login attempts and notify the account holder through their registered contact, requiring administrator intervention or a cool-down period before access is restored.

REQ - 2.1.5: The system shall enforce role-based access control so that each authenticated user type: establishment staff, BPLO administrator, OSCA administrator, PDAO administrator, and beneficiary can only access portal features and data corresponding to their assigned role.

2.2 Provisioning and Managing Establishment Accounts

Description and Priority

This feature enables BPLO officers to directly create, manage, and control iDetect Client Portal accounts for food service establishments in Valenzuela City. Rather than self-registering establishments, BPLO initiates account creation on behalf of eligible establishments, both newly registered businesses and existing ones upon verification of a valid Valenzuela City business permit. BPLO officers retain full authority to activate, suspend, or revoke establishment accounts at any time, without accessing beneficiary data or verification intelligence.

Priority: High

Stimulus / Response

Positive Flow

- A food service establishment (new or existing) approaches the BPLO office and requests iDetect access. The BPLO officer retrieves or verifies the establishment's business permit from the BPLO registry and initiates account creation on behalf of the establishment.

  Response: The system shall allow the BPLO officer to create an establishment account by entering the valid business permit number and designated staff credentials directly within the BPLO Portal, and shall immediately activate the account upon successful submission.

- The BPLO officer completes account creation and provides the establishment's staff with their login credentials.

  Response: The system shall generate and display the newly created account credentials to the BPLO officer, send an automated notification to the establishment (via email or SMS) containing their login details and instructions for accessing the Client Portal verification dashboard.

- An establishment staff initiates a password reset by entering their registered email or mobile number on the Client Portal login screen.

  Response: The system shall send a one-time reset link or PIN to the registered contact, allow the staff to set a new password upon verification, and log the reset event in the system audit trail.

Negative Flow

- A BPLO officer attempts to create an account using a business permit number that is not found in the BPLO registry, is expired, or does not correspond to a food service establishment in Valenzuela City.

  Response: The system shall block account creation, display a specific error message identifying the invalid or ineligible permit, and prevent the record from being saved until a valid permit is entered.

- A BPLO officer suspends or revokes an active establishment account due to a policy violation, business permit expiry, or closure.

  Response: The system shall immediately revoke the establishment's Client Portal access, notify the establishment staff of the suspension or revocation with a stated reason, and log the action including the BPLO officer's identity, timestamp, and reason in the audit trail.

- An establishment staff cannot complete self-service password reset because their registered contact details are outdated or inaccessible.

  Response: The system shall direct the staff to contact their BPLO officer, who may update the registered contact details and trigger a new reset through the BPLO Portal.

Functional Requirements

REQ - 2.2.1: The system shall allow a BPLO officer to directly create a Client Portal account for a food service establishment by entering a valid Valenzuela City business

permit number and designated staff credentials within the BPLO Portal, covering both newly registered and existing business establishments.

REQ - 2.2.2: The system shall validate the submitted business permit number against the BPLO registry in real time, and shall block account creation if the permit is not found, is expired, or belongs to a non-food-service establishment outside Valenzuela City.

REQ - 2.2.3: Upon account creation, the system shall automatically deliver login credentials and onboarding instructions to the establishment's registered contact (via email or SMS), and shall record the creating officer, timestamp, and associated permit number in the system log.

REQ - 2.2.4: The BPLO officer shall have authority to activate, suspend, or revoke any establishment account at any time through the BPLO administrative dashboard, without accessing flagged ID records, beneficiary personal information, or verification transaction records and the system shall send automated notifications to establishment staff upon any account status change (activation, suspension, revocation) initiated by a BPLO officer, including the stated reason for any adverse action.

REQ - 2.2.5: The system shall monitor the business permit expiry dates of all active establishment accounts and display a flagged alert on the BPLO Portal dashboard for any establishment whose permit is approaching expiry within 30 days or has already expired, without automatically suspending the account, so that the BPLO officer may take appropriate manual action.

REQ - 2.2.6: The system shall allow an establishment staff to reset their Client Portal password through a self-service mechanism by submitting their registered email address or mobile number, receiving a one-time reset credential, and setting a new password within a defined expiry window, with the event logged in the audit trail and The BPLO officer shall have authority to update an establishment's registered contact details and trigger a credential reset through the BPLO Portal when self-service password reset cannot be completed by the establishment staff.

REQ - 2.2.7: The BPLO Portal shall display an operational dashboard showing total active establishment accounts, recently provisioned accounts, pending administrative actions, and technical performance metrics to support BPLO oversight responsibilities without exposing any verification or beneficiary data.

2.3 Capturing and Preprocessing ID Images

Description and Priority

This feature allows establishment staff to capture a Senior Citizen or PWD ID image using the Client Portal mobile application camera. The system performs real-time image quality assessment and applies preprocessing steps: deskewing, contrast enhancement, and noise reduction to maximize OCR accuracy before extraction.

Priority: High

Stimulus / Response

Positive Flow

- An establishment staff opens the Client Portal and points the device camera at a Senior Citizen or PWD identification card.

  Response: The system shall display a real-time camera viewfinder with on-screen guidance indicating optimal ID positioning, lighting conditions, and distance, and provide instant visual feedback confirming image quality is sufficient before capturing.

- The establishment staff captures an image that passes the image quality assessment threshold.

  Response: The system shall automatically apply preprocessing algorithms including deskewing, contrast enhancement, and noise reduction to the captured image before passing it to the OCR extraction engine.

Negative Flow

- The establishment staff captures an image that is blurry, poorly lit, or significantly skewed beyond correctable limits.

  Response: The system shall display an image quality warning, reject the captured image, and prompt the staff to recapture the ID under better conditions before proceeding to OCR extraction.

- The establishment staff attempts to capture a physically damaged or heavily worn ID with illegible sections.

Response: The system shall notify the staff that image quality is insufficient for automated OCR processing and offer the manual data entry fallback option to continue the verification process.

Functional Requirements

REQ - 2.3.1: The system shall provide an establishment staff with a real-time camera viewfinder overlay displaying guidance on ID positioning, distance, and lighting to facilitate optimal image capture before OCR processing within the Client Portal mobile application.

REQ - 2.3.2: The system shall perform automated image quality assessment on each captured frame and reject images that fall below the defined quality threshold, prompting the establishment staff to recapture the ID before extraction proceeds.

REQ - 2.3.3: The system shall apply an automated preprocessing pipeline including deskewing, contrast enhancement, and noise reduction to all accepted captured ID images before passing them to the OCR engine to maximize extraction accuracy against non-standardized Philippine LGU credential formats.

REQ - 2.3.4: The system shall allow an establishment staff to switch to manual data entry as a fallback mechanism when OCR confidence scoring falls below the acceptable processing threshold, enabling continuation of the verification workflow without automated extraction.

REQ - 2.3.5: The Client Portal mobile application shall support Android 8.0 and later and iOS 12.0 and later to ensure compatibility with devices used across food service establishment environments in Valenzuela City.

2.4 Extracting and Validating OCR Data Fields

Description and Priority

This feature covers the automatic extraction of credential data from preprocessed ID images using OCR engine, applying confidence scoring and field-level pattern validation to verify that extracted data conforms to expected formats before presenting results for staff confirmation.

Priority: High

Stimulus / Response

Positive Flow

- The preprocessed ID image is passed to the OCR engine by the Client Portal.

  Response: The system shall automatically extract the ID number, full name, beneficiary type, validity period, and issuing office fields, apply confidence scoring to each extracted field, and display the extracted data on the staff's screen for confirmation before database verification.

- All extracted fields meet or exceed the confidence threshold and pass field-level pattern validation.

  Response: The system shall proceed automatically to real-time database verification using the extracted credential data without requiring additional staff input.

Negative Flow

- The OCR engine extracts data but one or more fields fall below the confidence threshold or fail field-level pattern validation.

  Response: The system shall highlight the low-confidence or invalid fields, retain extractable data in the form, and prompt the establishment staff to manually correct or confirm the flagged fields before proceeding to verification.

- The OCR engine is unable to extract any usable data from the preprocessed image.

  Response: The system shall display an extraction failure message and activate the manual data entry fallback, allowing the staff to type in the credential details directly.

Functional Requirements

REQ - 2.4.1: The system shall use OCR engine to automatically extract the ID number, full name, beneficiary type, validity period, and issuing office from preprocessed ID images, applying confidence scoring to each extracted field within the data extraction workflow.

REQ - 2.4.2: The system shall apply field-level pattern validation to all OCR-extracted data, verifying that ID number format, date ranges, and issuing office codes conform to expected Philippine LGU credential structures before proceeding to database verification.

REQ - 2.4.3: The system shall display OCR-extracted fields to the establishment staff for confirmation only when one or more fields carry a low confidence score or fail field-level pattern validation, highlighting the affected fields for manual review before proceeding to database verification. Fields that pass all confidence and validation checks shall not require additional staff confirmation.

REQ - 2.4.4: The system shall allow an establishment staff to manually edit any pre-populated field returned by the OCR engine before submitting data for database verification, ensuring staff can correct OCR errors without restarting the capture process.

2.5 Verifying Credentials Against Dual Databases

Description and Priority

This feature performs real-time simultaneous queries against both the OSCA and PDAO databases using the extracted credential data, returning a verification status: VALID, SUSPICIOUS, or INVALID to the establishment staff within 30 seconds without exposing comprehensive beneficiary personal data.

Priority: High

Stimulus / Response

Positive Flow

- The establishment staff confirms the OCR-extracted or manually entered credential data and initiates database verification.

Response: The system shall simultaneously query both the OSCA and PDAO databases through secure RESTful API integration and display the verification result: VALID, SUSPICIOUS, or INVALID with a clear visual indicator within 30 seconds from submission.

- The database query returns a VALID status for the submitted credential.

  Response: The system shall display a VALID result with green visual confirmation to the establishment staff and log the transaction in the audit history, without revealing any comprehensive personal data beyond the verification outcome.

Negative Flow

- The database query returns an INVALID status because the submitted ID number does not match any record in the OSCA or PDAO database.

  Response: The system shall display an INVALID result with a red visual indicator, inform the establishment staff of the non-match outcome, and prompt the option to flag the credential as suspicious for LGU investigation.

- The database query cannot be completed because the network connection is unavailable during the verification attempt.

  Response: The system shall display a connectivity error message, check the local cache for a recently verified matching record, and if

found, display a cached result with a timestamp of the last online verification.

- The OSCA and PDAO databases return conflicting results for the same submitted ID number for example, a record is found in one database but not the other.

  Response: The system shall return a SUSPICIOUS status to the establishment staff, log the conflict details in the audit trail, and automatically route a discrepancy alert to both the OSCA and PDAO portals for administrative review.

Functional Requirements

REQ - 2.5.1: The system shall simultaneously query both the OSCA and PDAO databases through secure RESTful API integration using the submitted ID number and beneficiary type to determine verification status within the dual-database verification engine.

REQ - 2.5.2: The system shall display the verification result using distinct color-coded visual indicators, green for VALID, amber for SUSPICIOUS, and red for INVALID to enable rapid result interpretation by establishment staff in high-volume point-of-sale environments.

REQ - 2.5.3: The system shall complete the full verification cycle from credential data submission to result display within 30 seconds under normal network conditions, meeting the real-time point-of-sale performance requirement.

REQ - 2.5.4: The system shall log each verification transaction including the credential data submitted, verification status returned, timestamp, and establishment identifier in the audit history database immediately upon completion of the verification cycle.

REQ - 2.5.5: The system shall return a SUSPICIOUS verification status and generate an automatic discrepancy alert routed to both the OSCA and PDAO portals when a submitted credential produces conflicting results across the two databases, logging the conflict details in the audit trail for administrative resolution.

REQ - 2.5.6: The system shall check local cache for recently verified matching records and display cached results with timestamp when network is unavailable.

2.6 Flagging and Reporting Suspicious Credentials

Description and Priority

This feature allows establishment staff to flag suspicious or invalid credentials with a single tap after a verification transaction. The system automatically routes the flag report including the captured ID image, extracted data, establishment identifier, and timestamp to the appropriate LGU portal for investigation.

Priority: High

Stimulus / Response

Positive Flow

- An establishment staff taps the flag button after receiving a SUSPICIOUS or INVALID verification result.

Response: The system shall display a confirmation prompt and, upon confirmation, transmit the captured ID image, OCR-extracted data, establishment identifier, transaction timestamp, and verification outcome to the OSCA Portal for Senior Citizen IDs or the PDAO Portal for PWD IDs.

- The flag report is successfully transmitted to the appropriate LGU portal.

  Response: The system shall display a confirmation message to the establishment staff, log the flag submission in the establishment's transaction history, and generate a reference number for tracking the report status.

Negative Flow

- An establishment staff attempts to submit a flag report while the device has no network connection.

  Response: The system shall queue the flag report locally on the device with all associated data and automatically transmit it to the appropriate LGU portal upon connectivity restoration, notifying the staff of the pending submission status.

- An establishment staff accesses the flag submission interface but decides not to proceed and cancels the action.

Response: The system shall discard the flag submission without transmitting any data to LGU portals and return the staff to the verification result screen without logging a flag event.

- An establishment staff attempts to flag a credential that returned INVALID because no matching record was found in either database, leaving the beneficiary type unconfirmed.

  Response: The system shall prompt the establishment staff to manually select whether the credential is a Senior Citizen ID (route to OSCA) or a PWD ID (route to PDAO) before allowing flag submission.

Functional Requirements

REQ - 2.6.1: The system shall allow an establishment staff to flag a suspicious credential with a single tap from the verification result screen, routing the report automatically to the OSCA Portal for Senior Citizen IDs or the PDAO Portal for PWD IDs based on the beneficiary type detected during verification.

REQ - 2.6.2: The system shall include the captured ID image, OCR-extracted data fields, establishment identifier, transaction timestamp, and verification outcome in every fraud flag transmission sent to the appropriate LGU portal.

REQ - 2.6.3: The system shall generate a unique reference number for each submitted flag report, display it to the establishment staff upon successful transmission, and associate it with the corresponding transaction entry in the audit history.

REQ - 2.6.4: The system shall queue fraud flag reports locally when the mobile device is offline and automatically submit them to the appropriate LGU portal upon connectivity restoration, preserving all associated data throughout the queuing period.

REQ - 2.6.5: The system shall prompt the establishment staff to manually select the beneficiary type: Senior Citizen or PWD before routing a flag report when the verification returned INVALID due to no record found in either database and the credential type could not be confirmed automatically.

REQ - 2.6.6: The system shall detect when a flag report is submitted for a credential that already has an active open investigation in the OSCA or PDAO portal, and shall log the new submission as a supplementary report linked to the existing investigation record rather than creating a duplicate case.

2.7 Managing Senior Citizen Records and Investigating Fraud (OSCA)

Description and Priority

This feature enables OSCA administrators to manage the Senior Citizen ID database, receive real-time notifications of flagged credentials with complete transaction context, review and investigate suspicious records, and mark confirmed fraudulent IDs to prevent future successful verifications across the network.

Priority: High

Stimulus / Response

Positive Flow

- A flag report for a Senior Citizen ID is transmitted from the Client Portal to the OSCA Portal.

  Response: The system shall immediately notify the OSCA administrator through the portal dashboard, displaying the captured ID image, extracted data fields, flagging establishment details, and transaction timestamp for review.

- An OSCA administrator reviews a flagged credential and confirms it as fraudulent.

  Response: The system shall update the credential status in the OSCA database to FRAUDULENT, propagate the status change in real time to all connected Client Portal instances, and prevent future VALID outcomes for that credential.

Negative Flow

- An OSCA administrator reviews a flagged credential and determines it is legitimate after investigation.

  Response: The system shall dismiss the flag, restore the VALID status in the database, and notify the establishment that submitted the flag of the investigation outcome.

- An OSCA administrator attempts to access PWD ID records or PDAO Portal data from within the OSCA Portal.

Response: The system shall deny access and display an authorization error, enforcing strict data compartmentalization that limits OSCA administrators to Senior Citizen records only.

Functional Requirements

REQ - 2.7.1: The OSCA Portal shall receive real-time notifications of each flag report transmitted from the Client Portal for Senior Citizen IDs, displaying the captured image, extracted data, flagging establishment identifier, and transaction timestamp within the OSCA investigation dashboard.

REQ - 2.7.2: The OSCA administrator shall investigate flagged credentials and mark confirmed fraudulent IDs as FRAUDULENT within the OSCA Portal, triggering an immediate real-time database status update that propagates across all connected Client Portal instances to prevent future VALID verifications of that credential.

REQ - 2.7.3: The system shall restrict OSCA Portal access to Senior Citizen ID records only, preventing OSCA administrators from viewing PWD records, PDAO portal data, or BPLO business account information at all times through enforced data compartmentalization.

REQ - 2.7.4: The OSCA Portal shall enable administrators to add, update, and manage Senior Citizen ID records in the OSCA database, including status changes reflecting investigation outcomes, renewal processing, and record corrections identified through dispute resolutions.

REQ - 2.7.5: The OSCA Portal shall generate fraud analytics reports displaying temporal patterns, geographic distribution of flagged credentials,

establishment-specific flagging trends, and flagged-to-confirmed conversion rates to support evidence-based enforcement strategies.

2.8 Managing PWD Records and Investigating Fraud (PDAO)

Description and Priority

This feature provides the PDAO Portal with parallel functionality to the OSCA Portal specifically for PWD credential management. PDAO administrators manage the PWD ID database, receive flagged credential reports, investigate suspicious records, and mark confirmed fraudulent IDs. PWD data is strictly compartmentalized from Senior Citizen records.

Priority: High

Stimulus / Response

Positive Flow

- A flag report for a PWD ID is transmitted from the Client Portal to the PDAO Portal.

  Response: The system shall immediately notify the PDAO administrator through the portal dashboard, displaying the captured ID image, extracted data fields, flagging establishment details, and transaction timestamp for investigation.

- A PDAO administrator reviews a flagged credential and confirms it as fraudulent.

Response: The system shall update the credential status in the PDAO database to FRAUDULENT, synchronize the status change in real time across all connected Client Portal instances, and generate an investigation outcome record for the audit trail.

Negative Flow

- A PDAO administrator reviews a flagged PWD credential and determines it is legitimate.

  Response: The system shall dismiss the flag, restore the VALID status in the PDAO database, and notify the flagging establishment of the resolution outcome.

- A PDAO administrator attempts to access Senior Citizen records or OSCA Portal data.

  Response: The system shall deny access and display an authorization error, enforcing strict data compartmentalization that limits PDAO administrators to PWD records only.

Functional Requirements

REQ - 2.7.1: The PDAO Portal shall receive real-time notifications of each flag report transmitted from the Client Portal for PWD IDs, displaying the captured image, extracted data, flagging establishment identifier, and transaction timestamp within the PDAO investigation dashboard.

REQ - 2.7.2: The PDAO administrator shall investigate flagged credentials and mark confirmed fraudulent IDs as FRAUDULENT within the PDAO Portal, triggering an

immediate real-time status update propagated across all connected Client Portal instances to prevent future VALID verifications of that credential.

REQ - 2.7.3: The system shall restrict PDAO Portal access to PWD ID records only, preventing PDAO administrators from viewing Senior Citizen records, OSCA portal data, or BPLO business account information through enforced data compartmentalization.

REQ - 2.7.4: The PDAO Portal shall enable administrators to add, update, and manage PWD ID records in the PDAO database, including status changes from investigation outcomes, renewal processing, and record corrections identified through beneficiary dispute resolutions.

REQ - 2.7.5: The PDAO Portal shall generate fraud analytics reports displaying temporal patterns, geographic distribution of flagged PWD credentials, establishment-specific trends, and flagged-to-confirmed conversion rates to support evidence-based enforcement and resource allocation.

2.9 Synchronizing Fraud Status Across Establishments

Description and Priority

This feature ensures that when an OSCA or PDAO administrator confirms a flagged credential as fraudulent, the updated FRAUDULENT status is immediately propagated to all connected Client Portal instances across all participating establishments, transforming isolated fraud detection into coordinated network-wide prevention.

Priority: High

Stimulus / Response

Positive Flow

- An OSCA or PDAO administrator marks a previously flagged credential as confirmed fraudulent in the portal.

  Response: The system shall immediately update the credential status in the master database and propagate the change in real time to all connected Client Portal instances, so that any subsequent verification attempt of that credential returns an INVALID result.

- An establishment staff attempts to verify a credential that was confirmed fraudulent after the last local cache refresh.

  Response: The system shall query the live database, detect the updated FRAUDULENT status, and return an INVALID verification result to the establishment staff.

Negative Flow

- A Client Portal instance is offline when OSCA or PDAO updates a credential status to FRAUDULENT.

  Response: The system shall synchronize the updated status to that Client Portal instance automatically upon connectivity restoration, ensuring the fraudulent credential cannot receive a VALID result once the device reconnects.

- An OSCA or PDAO administrator reverts a FRAUDULENT credential status to VALID following a successful dispute resolution.

Response: The system shall propagate the reinstated VALID status across all connected Client Portal instances in real time, allowing the credential to once again return a VALID outcome during verification transactions.

Functional Requirements

REQ - 2.9.1: The system shall propagate any credential status change confirmed by an OSCA or PDAO administrator in real time to all connected Client Portal instances, ensuring network-wide synchronization within 10 seconds of the database update under normal network conditions.

REQ - 2.9.2: The system shall return an INVALID verification result for any credential whose status has been updated to FRAUDULENT in the OSCA or PDAO master database, regardless of which establishment's Client Portal initiates the verification attempt.

REQ - 2.9.3: The system shall synchronize pending status updates to offline Client Portal instances automatically upon connectivity restoration to ensure fraudulent credential records are applied without requiring manual refresh by establishment staff.

REQ - 2.9.4: The system shall maintain a complete audit log of all credential status changes, including the administrator who made the change, the timestamp, the previous status, and the new status, for all OSCA and PDAO database modifications within the fraud synchronization workflow.

2.10 Registering a Beneficiary Account

Description and Priority

This feature allows Senior Citizens and PWDs to register accounts in the Beneficiary Mobile Application using their ID number and birthdate, verified against the OSCA or PDAO database and confirmed through an OTP sent to their registered mobile number. Registration is optional and does not affect a beneficiary's right to receive discounts.

Note: Beneficiary account deletion (right to erasure under RA 10173 Section 16) is acknowledged as a data subject right but is outside the scope of this capstone implementation. Beneficiaries seeking account removal are directed to their respective OSCA or PDAO office.

Priority: High

Stimulus / Response

Positive Flow

- A beneficiary enters their Senior Citizen or PWD ID number and birthdate in the registration screen of the Beneficiary Mobile Application.

  Response: The system shall verify the submitted ID number and birthdate against the corresponding OSCA or PDAO database, and if matched, send a one-time PIN to the mobile number registered with that credential for identity confirmation.

- The beneficiary enters the correct OTP within the allowed time window.

  Response: The system shall activate the beneficiary account, link it to the beneficiary's ID record in the OSCA or PDAO database, and navigate the beneficiary to their account dashboard.

Negative Flow

- The beneficiary enters an ID number and birthdate that do not match any record in the OSCA or PDAO database.

  Response: The system shall deny registration and display a message advising the beneficiary to verify their details with their local OSCA or PDAO office.

- The beneficiary fails to enter the OTP within the allowed time window.

  Response: The system shall expire the OTP, display a timeout notification, and offer the beneficiary the option to request a new OTP, with a defined maximum number of resend attempts per registration session.

Functional Requirements

REQ - 2.10.1: The system shall allow a beneficiary to initiate account registration by submitting their Senior Citizen or PWD ID number and birthdate, which the system verifies against the corresponding OSCA or PDAO database record before proceeding to OTP confirmation.

REQ - 2.10.2: The system shall send a one-time PIN to the mobile number registered in the beneficiary's OSCA or PDAO database record as the secondary verification factor during account registration, linking the beneficiary account to their registered device.

REQ - 2.10.3: The system shall activate a beneficiary account only upon successful validation of both the primary authentication factor (ID number and birthdate) and the secondary factor (OTP) within the registration workflow, ensuring account ownership is confirmed before access is granted.

REQ - 2.10.4: The system shall treat beneficiary account registration as optional, ensuring that business establishments provide mandated Senior Citizen and PWD discounts regardless of whether a beneficiary has registered in the application, creating no new technological barrier for vulnerable populations.

2.11 Sending Real-Time Transaction Notifications

Description and Priority

This feature delivers immediate SMS and push notifications to beneficiaries upon each verification attempt of their ID at participating establishments, providing the business name, transaction timestamp, and verification outcome including specific denial reasons, enabling beneficiaries to detect unauthorized credential use in near real time.

Priority: High

Stimulus / Response

Positive Flow

- An establishment staff completes a verification of a beneficiary's ID through the Client Portal.

  Response: The system shall immediately send both an SMS and a push notification to the beneficiary's registered mobile number and linked device, containing the establishment name, transaction date and time, and the verification outcome: Approved or Denied.

- A verification transaction results in a Denied outcome.

  Response: The system shall include the specific denial reason such as ID number mismatch, database record not found, ID expired, or status marked as FRAUDULENT in the notification delivered to the beneficiary.

Negative Flow

- A verification attempt occurs but the beneficiary's mobile number or device is temporarily unreachable at the time of the transaction.

  Response: The system shall log the transaction and queue the notification for delivery once the beneficiary's device is reachable, without delaying or disrupting the establishment's verification workflow.

- A beneficiary receives a transaction notification for an establishment they confirm they did not visit.

Response: The system shall provide a direct action within the notification interface allowing the beneficiary to submit an unauthorized use alert, which routes immediately to the appropriate OSCA or PDAO portal.

Functional Requirements

REQ - 2.11.1: The system shall deliver an SMS and push notification to the beneficiary's registered mobile number and linked device upon each verification transaction at a participating establishment, including the establishment name, transaction timestamp, and verification outcome (Approved or Denied).

REQ - 2.11.2: The system shall include the specific denial reason in the notification when a verification transaction returns a Denied outcome, providing the beneficiary with actionable information to understand the basis for the denial within the notification message.

REQ - 2.11.3: The system shall deliver SMS transaction notifications independently of mobile data access, ensuring that beneficiaries without active internet connectivity still receive verification alerts through the SMS channel.

REQ - 2.11.4: The system shall allow a beneficiary to submit an unauthorized credential use alert directly from within the transaction notification interface, automatically routing the alert to the appropriate OSCA or PDAO portal for immediate investigation.

REQ - 2.11.5: The system shall maintain a complete transaction notification history accessible within the Beneficiary Mobile Application, displaying all past verification

attempts associated with the beneficiary's ID in chronological order with establishment names, timestamps, and outcomes.

2.12 Checking ID Verification Status

Description and Priority

This feature allows authenticated beneficiaries to view their current ID verification status synchronized with OSCA or PDAO master databases, including last successful verification details, upcoming expiration dates, and issuing office contact information, enabling proactive verification of standing before discount transactions.

Priority: High

Stimulus / Response

Positive Flow

- An authenticated beneficiary navigates to the ID Status screen in the Beneficiary Mobile Application.

  Response: The system shall retrieve and display the beneficiary's current ID status: Active, Under Review, Suspended, or Expired, synchronized with the OSCA or PDAO master database, along with the last successful verification transaction details and upcoming expiration date.

- A beneficiary checks their status and finds it shows Active with no anomalies.

Response: The system shall display issuing office contact information alongside the status to allow the beneficiary to reach out proactively if they have questions about their record.

Negative Flow

- A beneficiary checks their ID status and finds it is marked Under Review or Suspended.

  Response: The system shall display the status with an explanation of what it means, provide the applicable dispute or appeal pathway if the beneficiary believes the status is incorrect, and display OSCA or PDAO contact details.

- The Beneficiary Mobile Application cannot retrieve the current ID status due to a network interruption.

  Response: The system shall display the last cached status with a timestamp of when it was last retrieved, and a notification that the displayed status may not reflect the latest database record until connectivity is restored.

12.3 Functional Requirements

REQ - 2.12.1: The system shall display the beneficiary's current ID status: Active, Under Review, Suspended, or Expired, synchronized with the OSCA or PDAO master database within the Beneficiary Mobile Application upon authenticated access to the ID Status screen.

REQ - 2.12.2: The system shall display the last successful verification transaction details including the establishment name, date, and outcome alongside the current ID status to provide the beneficiary with recent verification activity context.

REQ - 2.12.3: The system shall display the beneficiary's ID expiration date and issuing office contact information within the ID Status screen to enable proactive renewal and direct communication with the issuing authority.

REQ - 2.12.4: The system shall display the last cached ID status with a retrieval timestamp when a beneficiary accesses the ID Status screen without network connectivity, clearly indicating that the displayed information may not reflect the current database record.

2.13 Submitting and Tracking Dispute Resolutions

Description and Priority

This feature provides Senior Citizens and PWDs with a structured dispute mechanism to challenge erroneous verification denials. Beneficiaries select a dispute category, provide a written explanation, and optionally upload supporting documentation. The system routes the dispute to OSCA or PDAO for human-reviewed adjudication with guaranteed acknowledgment and resolution timelines.

Priority: High

Stimulus / Response

Positive Flow

- A beneficiary selects a denied transaction from their notification history and taps the Dispute button.

  Response: The system shall open the dispute submission form pre-populated with the transaction reference number, establishment details, and denial reason, and present the dispute category selection menu to the beneficiary.

- The beneficiary completes the dispute form by selecting a category, entering a written explanation, and optionally uploading supporting documentation, then submits the dispute.

  Response: The system shall route the dispute to the appropriate OSCA or PDAO investigation queue, generate a dispute reference number, and send an automatic acknowledgment notification to the beneficiary within 24 hours confirming receipt.

Negative Flow

- The beneficiary attempts to submit a dispute form without providing a written explanation in the required field.

  Response: The system shall display a validation error on the explanation field and prevent submission until the mandatory content is provided, with an on-screen message indicating why the form cannot be submitted.

- The OSCA or PDAO dispute review exceeds the service-level agreement timeframe without resolution.

Response: The system shall escalate the overdue dispute within the LGU portal, send a status update notification to the beneficiary confirming the case is still under active review, and record the escalation event in the audit trail.

Functional Requirements

REQ - 2.13.1: The system shall allow a beneficiary to initiate a dispute from a denied transaction record in the notification history, pre-populating the dispute form with the transaction reference number, establishment details, and denial reason to minimize manual input.

REQ - 2.13.2: The system shall present the beneficiary with three dispute category options: Wrong Denial (I Am Valid), ID Used Without My Authorization, or Database Contains Incorrect Information, as the first step of the dispute submission workflow. Business discount refusal incidents shall be submitted separately through the fraud reporting feature in Section 14.

REQ - 2.13.3: The system shall require a beneficiary to provide a written explanation in the dispute form and allow the optional upload of supporting documentation such as physical ID photos or barangay certifications before the dispute submission can proceed.

REQ - 2.13.4: The system shall route the submitted dispute to the appropriate OSCA or PDAO investigation queue based on the beneficiary's ID type, send an automatic acknowledgment notification within 24 hours of submission, and provide a unique dispute reference number for tracking.

REQ - 2.13.5: The system shall enable the beneficiary to track the status of a submitted dispute from within the application, displaying status updates as OSCA or PDAO staff review evidence, and delivering a final resolution notification with explanation upon case closure.

REQ - 2.13.6: The OSCA or PDAO staff shall judge all disputes through human review within the portal, with the system enforcing service-level agreements of 24-hour acknowledgment and 48-to-72-hour resolution for database errors, and automated escalation notifications for cases exceeding defined timelines.

2.14 Reporting Fraud as a Beneficiary

Description and Priority

This feature enables beneficiaries to report three types of fraud incidents that businesses and government agencies cannot independently detect: unauthorized credential use by another person, business refusal of a system-approved discount, and database discrepancy alerts for incorrect personal information in their displayed records.

Priority: High

Stimulus / Response

Positive Flow

- A beneficiary receives a transaction notification for an establishment they confirm they did not visit and submits an unauthorized use alert.

Response: The system shall route the alert immediately to the appropriate OSCA or PDAO portal with the transaction details, prompt the administrator to investigate, and if confirmed unauthorized, flag the credential for cross-establishment network protection.

- A beneficiary submits a business refusal report indicating that an establishment denied their discount despite the system returning a VALID result.

  Response: The system shall route the report to the BPLO Portal for investigation of a potential DOJ Opinion violation, log the incident with transaction details, and send an acknowledgment notification to the beneficiary.

- A beneficiary identifies incorrect personal information in their displayed ID record and submits a database discrepancy alert.

  Response: The system shall route the discrepancy report to the appropriate OSCA or PDAO portal for verification and correction, acknowledge receipt to the beneficiary with a reference number, and update the beneficiary's record upon confirmation by the LGU administrator.

Negative Flow

- A beneficiary submits a fraud report without providing sufficient details to enable investigation.

Response: The system shall display a validation prompt requesting minimum required information before allowing submission, and provide guidance on what details are needed for the report to be actionable by LGU investigators.

Functional Requirements

REQ - 2.14.1: The system shall allow a beneficiary to submit an unauthorized credential use alert when their ID is used at an establishment they did not visit, routing the alert immediately to the appropriate OSCA or PDAO portal with the transaction reference, establishment details, and beneficiary confirmation statement.

REQ - 2.14.2: The system shall allow a beneficiary to submit a business refusal report when an establishment denies a discount despite the system returning a VALID verification result, routing the report to the BPLO Portal for investigation of potential DOJ Opinion non-compliance. The BPLO Portal shall receive and display beneficiary-submitted business refusal reports routed from the Beneficiary Mobile Application, showing the transaction reference, establishment identifier, verification outcome, and beneficiary confirmation statement, enabling the BPLO officer to investigate potential DOJ Opinion non-compliance by the flagged establishment.

REQ - 2.14.3: The system shall allow a beneficiary to submit a database discrepancy alert when they identify incorrect personal information in their displayed record, routing the report to the appropriate OSCA or PDAO portal for verification and correction of the master database entry.

REQ - 2.14.4: The system shall generate an acknowledgment notification with a reference number for each beneficiary-submitted fraud report, confirming receipt and providing the beneficiary with a means to track the investigation status within the application.

2.15 Operating Offline and Synchronizing Data

Description and Priority

This feature provides limited offline capability for the Client Portal by caching recently verified credentials locally, and allows the Beneficiary Mobile Application to display previously loaded information when network access is unavailable. Pending actions queue locally and synchronize automatically upon connectivity restoration.

Priority: Medium

Stimulus / Response

Positive Flow

- An establishment staff attempts a verification while the device has no network connection.

  Response: The system shall check the local cache for a recently verified matching record and, if found, display the cached result with a timestamp indicating when the credential was last verified online and a clear label that this is a cached offline result.

- The mobile device reconnects to the network after a period of offline operation.

Response: The system shall automatically synchronize all queued fraud flag reports and pending actions with the server, and update the local cache with the latest database verification statuses without requiring manual intervention from the staff.

Negative Flow

- An establishment staff attempts to verify an ID not present in the local cache while the device is offline.

  Response: The system shall display an offline mode warning stating that real-time verification is unavailable, advise the staff to defer the transaction or proceed provisionally in compliance with DOJ provisional approval requirements, and prompt for reconnection.

- The device fails to synchronize queued reports after connectivity restoration due to a server-side error.

  Response: The system shall retain the queued reports locally, retry synchronization automatically at the next available opportunity, and notify the staff of the pending synchronization status through a persistent indicator in the interface.

Functional Requirements

REQ - 2.15.1: The system shall cache recently verified credentials locally on the Client Portal mobile device to enable limited offline verification capability, displaying cached results with clear offline mode labeling and the timestamp of the last online verification to the establishment staff.

REQ - 2.15.2: The system shall queue fraud flag reports and other pending actions locally when the Client Portal device is offline and automatically submit them to the appropriate LGU portal upon connectivity restoration without requiring manual resubmission by staff.

REQ - 2.15.3: The system shall display a persistent offline mode indicator to the establishment staff and beneficiary whenever real-time database verification is unavailable due to network interruption, distinguishing cached results from live verification outcomes throughout the offline session.

REQ - 2.15.4: The Beneficiary Mobile Application shall allow beneficiaries to view previously loaded transaction history and ID status information when network access is unavailable, with a clear indication that the displayed data reflects the last synchronized state and may not be current.

2.16 Logging Audit Trails and Transaction History

Description and Priority

This feature ensures that all verification transactions, fraud flag submissions, database status changes, administrator actions, and dispute events are permanently logged in a tamper-evident audit trail stored in the system database, supporting compliance reporting, accountability tracking, and fraud pattern analysis.

Priority: High

Stimulus / Response

Positive Flow

- A verification transaction is completed through the Client Portal.

  Response: The system shall immediately log the transaction record in the audit database including the credential data submitted, verification status returned, establishment identifier, staff account identifier, and timestamp.

- An OSCA or PDAO administrator marks a credential as fraudulent.

  Response: The system shall log the action in the audit trail with the administrator's account identifier, the credential affected, the previous and new status, and the exact timestamp, creating a permanent non-repudiable record of the database modification.

- An establishment staff requests a compliance report summary for a defined date range from the transaction history screen.

  Response: The system shall generate a summary report of all verification transactions within the requested period, including counts of VALID, SUSPICIOUS, and INVALID outcomes, and present it on screen for staff review.

Negative Flow

- An user attempts to delete or modify a logged audit record.

  Response: The system shall deny the modification, maintain the integrity of all audit records, and log the attempted modification as a security event in the audit trail.

Functional Requirements

REQ - 2.16.1: The system shall permanently log every verification transaction in the audit database immediately upon completion, including the credential data submitted, verification status, establishment identifier, staff account identifier, and transaction timestamp to support accountability tracking.

REQ - 2.16.2: The system shall log all database status changes made by OSCA or PDAO administrators in the audit trail, recording the administrator account identifier, the affected credential, the previous status, the new status, and the exact timestamp of each modification.

REQ - 2.16.3: The system shall log all fraud flag submissions, dispute events, and administrator investigation outcomes in the audit trail with associated reference numbers, timestamps, and user identifiers to maintain complete traceability across the fraud management lifecycle.

REQ - 2.16.4: The system shall prevent any user from deleting or modifying audit log records, maintaining tamper-evident transaction history that supports regulatory compliance, fraud investigation, and legal proceedings when required.

REQ - 2.16.5: The system shall allow establishment staff to generate compliance report summaries from their transaction history for defined date ranges, displaying counts and outcomes of verification transactions to support internal audit and regulatory reporting requirements.

REQ - 2.16.6: The system shall retain all audit log records for a minimum of 5 years from the date of creation, after which records may be archived to long-term storage,

supporting regulatory compliance, fraud investigation, and legal evidentiary requirements under applicable Philippine data governance laws.

REQ - 2.16.6: The system shall restrict audit trail read access to BPLO administrators for business account and system operation logs, OSCA administrators for Senior Citizen verification and fraud investigation logs, and PDAO administrators for PWD verification and fraud investigation logs, in accordance with the data compartmentalization rules enforced across all portals.

2.17 Securing Data and Ensuring Privacy Compliance

Description and Priority

This feature covers the security controls, encrypted data transmission, data minimization principles, and Republic Act 10173 (Data Privacy Act of 2012) compliance mechanisms implemented across all iDetect portals, ensuring beneficiary personal information is protected while enabling transparency and access rights for data subjects.

Priority: High

Stimulus / Response

Positive Flow

- An establishment staff submits a verification transaction that queries the OSCA or PDAO database.

  Response: The system shall transmit all data over encrypted channels using industry-standard protocols, return only the binary verification

status to the Client Portal without exposing comprehensive personal data, and store only the minimum necessary transaction data in the audit log.

- A beneficiary exercises their RA 10173 Section 16 right to access their verification records.

  Response: The system shall provide the beneficiary with access to their current ID status, transaction history, and any dispute records associated with their credential through the authenticated Beneficiary Mobile Application, fulfilling the statutory data access right.

Negative Flow

- An unauthorized actor attempts to access the OSCA or PDAO database directly through an unauthenticated API request.

  Response: The system shall reject the request, log the unauthorized access attempt with the request source and timestamp in the security audit trail, and notify the system administrator of the security event.

- A user attempts to access another user role's data or portal features beyond their assigned permissions.

  Response: The system shall deny access, display an authorization error message, and log the unauthorized access attempt in the audit trail for administrator review.

Functional Requirements

REQ - 2.17.1: The system shall transmit all data between the Client Portal, LGU portals, and backend databases over encrypted channels using industry-standard encryption protocols, protecting credential data and verification results from interception during transmission.

REQ - 2.17.2: The system shall implement role-based access control across all portals ensuring that BPLO administrators access only business account data, OSCA administrators access only Senior Citizen records, PDAO administrators access only PWD records, establishment staff access only verification results, and beneficiaries access only their own records.

REQ - 2.17.3: The system shall return only binary verification status responses: VALID, SUSPICIOUS, or INVALID to the Client Portal without transmitting comprehensive beneficiary personal data, implementing privacy-preserving query design that minimizes data exposure to business establishments.

REQ - 2.17.4: The system shall provide beneficiaries with access to their verification status, transaction history, and dispute records through the authenticated Beneficiary Mobile Application, fulfilling the data subject access rights mandated by Republic Act 10173 Section 16 of the Data Privacy Act of 2012.

REQ - 2.17.5: The system shall log all unauthorized access attempts, authentication failures, and permission violation events in a dedicated security audit trail with the source, timestamp, and type of event, supporting security monitoring and incident response across all portals.

REQ - 2.17.6: The system shall support generation of a breach incident report by the system administrator in the event of a confirmed data security breach, containing the nature of the breach, affected data categories, and estimated number of affected data subjects, in accordance with the 72-hour notification obligation to the National Privacy Commission under Republic Act 10173.

REQ - 2.17.7: The Client Portal mobile application and Beneficiary Mobile Application shall enforce automatic session expiry after a defined period of inactivity, requiring the user to re-authenticate before resuming access, to prevent unauthorized use of unattended authenticated sessions on shared or lost devices.

2.18 Designing an Accessible and Multilingual Interface

Description and Priority

This feature ensures all iDetect interfaces employ large touch targets, high-contrast color schemes, bilingual support in English and Filipino, minimal text input requirements, and progressive disclosure of complex features, maintaining WCAG 2.1 AA accessibility standards to accommodate users with varying technical literacy and abilities.

Priority: Medium

Stimulus / Response

Positive Flow

- A beneficiary aged 60 or above opens the Beneficiary Mobile Application for the first time.

Response: The system shall display large touch targets, high-contrast text, clear iconography, and simple onboarding guidance in both English and Filipino, minimizing cognitive load for elderly users with potentially lower digital literacy.

- An establishment staff with no prior training opens the Client Portal on their first day.

  Response: The system shall guide the staff through the three-step verification workflow: launch, scan, view result using inline contextual prompts and clear visual feedback at each step, enabling proficiency without prior formal training.

Negative Flow

- A beneficiary switches the application language preference from English to Filipino.

  Response: The system shall immediately render all interface labels, instructions, notifications, and status messages in Filipino throughout the application without requiring a session restart.

- A user interacts with a feature beyond the primary verification or status view workflow.

  Response: The system shall apply progressive disclosure, surfacing only the information and controls needed for the immediate task and revealing advanced features only upon deliberate navigation.

Functional Requirements

REQ - 2.18.1: The Client Portal and Beneficiary Mobile Application shall employ large touch targets meeting WCAG 2.1 AA minimum size guidelines and high-contrast color schemes to accommodate users with varying visual acuity and motor dexterity, including elderly Senior Citizen beneficiaries.

REQ - 2.18.2: All iDetect portals and mobile applications shall provide full bilingual support in English and Filipino, allowing users to switch language preference at any time with all interface text, notifications, and guidance rendered in the selected language immediately upon change.

REQ - 2.18.3: The Client Portal verification workflow shall adhere to the three-action limit of launch, scan, and view result by implementing automatic background authentication, immediate camera activation upon launch, and automatic processing without intermediate confirmation dialogs to minimize adoption resistance.

REQ - 2.18.4: The system shall apply progressive disclosure across all portals and applications, surfacing only the controls and information required for the immediate task during primary workflows and revealing advanced features such as fraud analytics dashboards and bulk management tools only upon deliberate navigation.

REQ - 2.18.5: All iDetect interfaces shall minimize mandatory text input requirements, pre-populating fields where data is already known from the authenticated session or the current transaction context, to reduce burden on frontline establishment staff and elderly or low-literacy beneficiaries.